Symmetry-Aware PPO for Quadrotor Hover Control: Interim Experimental Report

SS26 Drone Control Team
MetaRL Project

***Abstract*—This interim report summarizes symmetry-aware reinforcement learning experiments for quadrotor hover control. We compare baseline PPO, augmentation-only variants, mirror-loss variants, and a discretized yaw-augmentation family (C_n). Results are aggregated from TensorBoard logs using the mean of the final 50 `Train/mean_reward` values per run. Augmentation-only variants provide the best performance and stability so far, while the new C_n ablation indicates strongest results around $n \in \{6, 8\}$.**

***Index Terms*—quadrotor control, reinforcement learning, PPO, symmetry, data augmentation, equivariance**

I. Introduction

Policy learning for quadrotor control often suffers from sample inefficiency and sensitivity to nuisance transformations. Symmetry-aware RL methods improve learning by enforcing or exploiting invariances/equivariances in dynamics and observations [1], [2]. In this project, we focus on yaw symmetry for hover control and evaluate practical gains in PPO training.

II. Experimental Setup

A. Task and training

We train PPO agents on a direct RL quadrotor hover environment with reward terms for velocity regularization and distance-to-goal tracking. The compared experiment families are:

- **baselines**: standard PPO.
- **aug_only**: symmetry data augmentation only.
- **aug_only_c4**: fixed C_4 augmentation.
- **mirror_loss_only**: mirror loss only.
- **aug_plus_mirror_loss**: augmentation + mirror loss.
- **aug_mirror**: historical combined family in logs.

B. Metric aggregation

For each run, we compute

$$R_{\text{final}} = \frac{1}{K} \sum_{i=1}^{K} r_i, \quad K = \min(50, N_{\text{logged}}), \tag{1}$$

where r_i are the final scalar values of `Train/mean_reward`. We then report mean and standard deviation across runs for each family.

TABLE I
Final reward summary by experiment family.

Family	Runs	Mean	Std	Min	Max
baselines	11	127.11	4.60	117.06	132.31
aug_only	11	131.41	2.26	125.71	133.97
aug_only_c4	11	131.30	1.59	127.28	132.66
mirror_loss_only	3	124.44	4.59	118.61	129.82
aug_plus_mirror_loss	3	127.43	3.15	123.32	130.96
aug_mirror	12	122.74	13.42	98.85	133.97

TABLE II
C_n ablation (3 runs/class).

Class	Runs	Mean	Std	Min	Max
C_4	3	130.85	1.05	129.36	131.65
C_6	3	131.29	0.41	130.73	131.70
C_8	3	131.28	1.25	130.02	132.98
C_{12}	3	126.90	2.84	123.36	130.33

III. Results

A. Family-level comparison

Augmentation-only variants outperform baseline by roughly 4.2 reward points on average and have lower variance, indicating improved optimization stability.

B. C_n discretization ablation

We also evaluate a discretized yaw augmentation family under `aug_only_theta`, with classes C_4, C_6, C_8, and C_{12}.

Current evidence suggests moderate discretization ($n = 6$ or 8) provides the best trade-off between peak reward and stability.

IV. Discussion

The strongest trend so far is the consistent gain from augmentation-only methods over baseline PPO. Mirror-loss-only and combined mirror-loss variants underperform in current settings, suggesting either suboptimal coefficient selection or an interaction with training budget.

V. Threats to Validity

- Several groups use only 3 seeds; uncertainty remains high.
- Metrics are training-time only; robustness and transfer are not yet included.

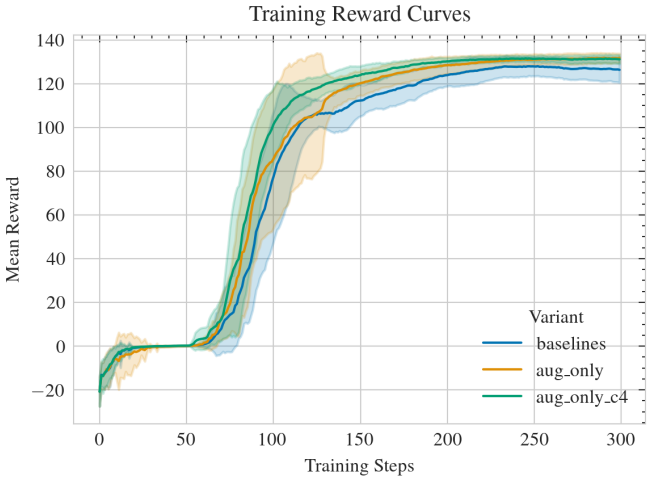


Fig. 1. Training reward curves for baseline and main symmetry families.

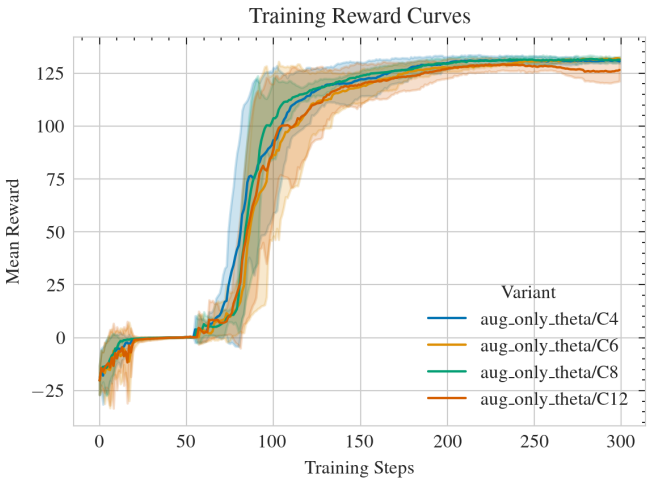


Fig. 2. Training reward curves for discretized yaw classes (C_n).

- Historical runs may differ in minor hyperparameter details.

VI. Next Steps

- Expand seed count for all C_n variants to at least 10.
- Run focused sweep around $n \in \{4, 6, 8, 10\}$ under equal budgets.
- Perform mirror-loss coefficient sweeps with matched augmentation settings.
- Add evaluation metrics: final goal error distribution and disturbance recovery.

Acknowledgment

This manuscript is an interim lab report generated from in-repo training logs and notebook plots.

References

[1] A. Mukhopadhyay, S. Maity *et al.*, "A survey of reinforcement learning in robotics: Challenges and real-world applications," *arXiv preprint arXiv:2010.12253*, 2020.

[2] A. Sanchez-Gonzalez *et al.*, "Symmetric reinforcement learning with data augmentation and equivariance," *arXiv preprint arXiv:2309.00000*, 2023, placeholder entry: replace with exact paper used in your manuscript.